

Spring 2025 MC Problem 2: WARP Violation

ECO 310

1 The Problem

On Monday, Homer Simpson buys a bundle of food and drink that costs \$90. On Tuesday, he buys a different bundle, for \$100, that would have cost \$80 at the Monday prices. In which of the following situations does he violate WARP:

- (a) If the Monday bundle would cost \$100 at the Tuesday prices.
- (b) If the Monday bundle would cost more than \$100 at the Tuesday prices.
- (c) Homer does not violate WARP.

Answer: (a)

2 Setting Up the Problem

Let's define our notation clearly:

- p^M = Monday prices (price vector)
- p^T = Tuesday prices (price vector)
- x^M = Monday bundle (what Homer bought Monday)
- x^T = Tuesday bundle (what Homer bought Tuesday)

2.1 What We Know

From the problem statement:

$$\begin{array}{ll} p^M \cdot x^M = 90 & \text{(Monday bundle cost on Monday)} \\ p^M \cdot x^T = 80 & \text{(Tuesday bundle cost at Monday prices)} \\ p^T \cdot x^T = 100 & \text{(Tuesday bundle cost on Tuesday)} \end{array}$$

We need to determine what happens when $p^T \cdot x^M$ (Monday bundle cost at Tuesday prices) takes different values.

3 Understanding WARP

Weak Axiom of Revealed Preference (WARP): If bundle x is chosen when bundle y is affordable, then whenever y is chosen, x must not be affordable.

More formally: If x is directly revealed preferred to y , then y cannot be directly revealed preferred to x .

3.1 Mathematical Statement

If at prices p and income m :

$$p \cdot x \leq m \quad \text{and} \quad p \cdot y \leq m \quad \text{and we choose } x$$

Then x is **revealed preferred** to y (denoted $x \succsim^R y$).

WARP says: If $x \succsim^R y$, then we cannot also have $y \succsim^R x$ (unless $x = y$).

4 Applying WARP to Homer's Problem

4.1 Monday's Revealed Preference

On Monday, Homer had budget \$90. He:

- **Chose** bundle x^M at cost $p^M \cdot x^M = 90$
- **Could afford** bundle x^T at cost $p^M \cdot x^T = 80 < 90$

Since Homer chose x^M when x^T was strictly cheaper (and thus affordable), this reveals:

$$\boxed{x^M \succ x^T} \quad (\text{Monday bundle is **strictly** preferred to Tuesday bundle})$$

This is a **strict** preference because he chose x^M and didn't spend his full budget on x^T .

4.2 Tuesday's Revealed Preference

On Tuesday, Homer had budget \$100. He:

- **Chose** bundle x^T at cost $p^T \cdot x^T = 100$
- Bundle x^M costs $p^T \cdot x^M = ?$

Case (a): If $p^T \cdot x^M = 100$

Homer could afford x^M (costs exactly his budget), but chose x^T . This reveals:

$$x^T \succsim x^M \quad (\text{Tuesday bundle is weakly preferred to Monday bundle})$$

This is a WARP violation! We have:

- Monday revealed: $x^M \succ x^T$ (strict preference for Monday bundle)
- Tuesday revealed: $x^T \succsim x^M$ (weak preference for Tuesday bundle)

These preferences are **inconsistent**.

Case (b): If $p^T \cdot x^M > 100$

Homer **cannot afford** x^M on Tuesday. Since x^M is not in his budget set, we cannot infer anything about his preferences from Tuesday's choice. Therefore, **no WARP violation**.

5 The WARP Violation Inequality

Homer violates WARP when **both** of the following hold:

$$p^M \cdot x^T < p^M \cdot x^M \quad (\text{Monday reveals } x^M \succ x^T)$$

$$p^T \cdot x^M \leq p^T \cdot x^T \quad (\text{Tuesday reveals } x^T \succsim x^M)$$

In our specific problem:

$$80 < 90 \quad \checkmark \quad (\text{Monday condition satisfied})$$

$$p^T \cdot x^M \leq 100 \quad (\text{Tuesday condition - when does this hold?})$$

Answer: WARP is violated when $p^T \cdot x^M \leq 100$

This includes option (a) where $p^T \cdot x^M = 100$, but **not** option (b) where $p^T \cdot x^M > 100$.

6 Intuition

Think of it this way:

- On Monday, Homer had a choice between the two bundles (both affordable) and picked Monday's bundle. This shows he prefers it.
- On Tuesday, if the Monday bundle is still affordable (costs $\leq \$100$), but Homer chooses the Tuesday bundle instead, he's **contradicting** his Monday preference.
- However, if the Monday bundle becomes too expensive on Tuesday (costs $> \$100$), then Homer simply can't afford it anymore. Choosing the Tuesday bundle doesn't contradict anything because he had no choice.

7 General WARP Formula

For any two observations (p^1, x^1) and (p^2, x^2) , WARP is violated if:

$$\boxed{\begin{cases} p^1 \cdot x^2 \leq p^1 \cdot x^1 & (\text{bundle 2 was affordable when we chose bundle 1}) \\ p^2 \cdot x^1 \leq p^2 \cdot x^2 & (\text{bundle 1 was affordable when we chose bundle 2}) \\ x^1 \neq x^2 & (\text{the bundles are different}) \end{cases}}$$

with at least one inequality strict.

8 Summary

Homer violates WARP in option (a) because:

- Monday: Chose x^M over affordable $x^T \Rightarrow x^M \succ x^T$
- Tuesday: Chose x^T over affordable $x^M \Rightarrow x^T \succsim x^M$
- These preferences contradict each other

Homer does **not** violate WARP in option (b) because x^M is not affordable on Tuesday, so no preference reversal is revealed.